

# An effective lateral transshipment policy to improve service level in the supply chain

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## Abstract

A new lateral transshipment policy, called service level adjustment (SLA), is proposed. SLA differs from previous policies by integrating emergency lateral transshipment with preventive lateral transshipment to efficiently respond to customer demands. Additionally, the proposed policy considers the service level to decide the quantity for lateral transshipment. Experiments simulating a two-echelon supply chain were conducted. The proposed policy has lower total costs than previous transshipment policies. Moreover, our policy more effectively responds to changes in demands, penalty cost, and ordering cost.

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**Keywords:** Supply chain; Lateral transshipment; Service level

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## 1. Introduction

Retailers in the supply chain face great uncertainty regarding consumer demands. If retailers that require long replenishment lead times from suppliers are located closer to each other or spend significant funds on construction and operation of storage facilities to prevent stockout, a lateral transshipment policy can be used as an effective alternative to minimizing total cost. Lateral transshipment is defined as the redistribution of stock from retailers with stock on hand to retailers that cannot meet customer demands or to retailers that expect significant losses due to high risk (Tagaras, 1999). Although transportation cost is increased, lateral transshipment is known as a better approach than a policy of no transshipments (Burton and Banerjee, 2005).

Lateral transshipment has mainly been applied to supply the military or the airline company with airplane parts. Wong et al. (2005) dealt with an analytical model to obtain the optimal spare parts stocking levels in a single-item, multi-hub, multi-company, repairable inventory system in which complete pooling of stock is permitted among the hubs and companies. This study was motivated by inspection of airplanes in hub airports. However, in the other industries, the execution of lateral transshipment has been difficult due to information extinction and insufficient understanding between companies or departments. However, large

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|                              |                                                                                                 |                                |                                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nomenclature</b>          |                                                                                                 | SO <sub>it</sub>               | quantity for stockout of retailer <i>i</i> at period <i>t</i>                                                                                                 |
| <i>Indices</i>               |                                                                                                 | BO <sub>it</sub>               | quantity for backorder of retailer <i>i</i> at the end of period <i>t</i>                                                                                     |
| <i>i</i>                     | index for retailers                                                                             | SLRP <sub>iRP(t)</sub>         | probability that stockout does not occur for retailer <i>i</i> in RP( <i>t</i> )                                                                              |
| <i>j</i>                     | index for retailers with more stock than desired                                                | CH                             | holding cost per unit stock in the unit period for any retailers                                                                                              |
| <i>k</i>                     | index for retailers with less stock than desired                                                | CP                             | penalty cost per unit stock in the unit period for any retailers                                                                                              |
| <i>t</i>                     | index for periods                                                                               | CTQ                            | transportation cost per unit stock                                                                                                                            |
| <i>Sets</i>                  |                                                                                                 | CTO                            | ordering cost per order                                                                                                                                       |
| <i>I</i>                     | set of retailers, $I = \{i   i = 1, \dots, n\}$                                                 | RSs <sub>it</sub>              | quantity for lateral transshipment that retailer <i>i</i> can take from other retailers at the end of period <i>t</i>                                         |
| <i>J</i>                     | set of retailers with more stock than desired, $J \subset I$                                    | ERS <sub>it</sub>              | required quantity for lateral transshipment except OH <sub>it</sub> from transshipment target stock level for retailer <i>i</i> at the end of period <i>t</i> |
| <i>K</i>                     | set of retailers with less stock than desired, $K \subset I, J \cap K = \phi$                   | AS <sub>it</sub>               | available quantity that retailer <i>i</i> can supply at the end of period <i>t</i>                                                                            |
| <i>Notations</i>             |                                                                                                 | $\alpha$                       | upper level of SLRP                                                                                                                                           |
| <i>R</i>                     | review period                                                                                   | $\beta$                        | target level of SLRP                                                                                                                                          |
| <i>L</i>                     | lead time for lateral transshipment from a supplier to retailers                                | $\gamma$                       | lower level of SLRP                                                                                                                                           |
| <i>N</i>                     | number of retailers                                                                             | $\Phi(\cdot)$                  | probability density function of standard normal distribution with mean 0 and variance 1                                                                       |
| RP( <i>t</i> )               | remaining period until reorder point after the period <i>t</i>                                  | $z_\alpha$                     | <i>z</i> -value with proportion $\alpha$ of the area under the normal curve                                                                                   |
| <i>D</i> <sub>it</sub>       | demand of a retailer <i>i</i> at period <i>t</i>                                                | $z_\beta$                      | <i>z</i> -value with proportion $\beta$ of the area under the normal curve                                                                                    |
| SD <sub>iRP(t)</sub>         | demand of a retailer <i>i</i> during RP( <i>t</i> ),<br>$SD_{iRP(t)} = \sum_{t'=t+1}^R D_{it'}$ | int( $\cdot$ )                 | function to return the integer value                                                                                                                          |
| $\mu(SD_{iRP(t)})$           | mean of demand of retailer <i>i</i> during RP( <i>t</i> )                                       | <i>Decision variables</i>      |                                                                                                                                                               |
| $\sigma(SD_{iRP(t)})$        | standard deviation of demand of retailer <i>i</i> during                                        | <i>X</i> <sub>jk<i>t</i></sub> | quantity of lateral transshipment from retailer <i>j</i> to retailer <i>k</i> at period <i>t</i>                                                              |
| <i>S</i> <sub><i>i</i></sub> | order-up-to level of retailer <i>i</i>                                                          | <i>e</i> <sub>jk<i>t</i></sub> | 1: If lateral transshipment is performed between retailer <i>j</i> and retailer <i>k</i> at period <i>t</i>                                                   |
| <i>Q</i> <sub>it</sub>       | order quantity for stock of retailer <i>i</i> at the end of period <i>t</i>                     |                                |                                                                                                                                                               |
| INV <sub>it</sub>            | inventory of retailer <i>i</i> at period <i>t</i>                                               |                                |                                                                                                                                                               |
| OH <sub>it</sub>             | on-hand stock of retailer <i>i</i> at period <i>t</i>                                           |                                |                                                                                                                                                               |

modern companies connected by information systems can control the relationships of many branches, and thus they may be poised to reap cost reduction and service improvement associated with lateral transshipment.

Lateral transshipment can be divided into emergency lateral transshipment (ELT) and preventive lateral transshipment (PLT). ELT mandates emergency redistribution from a retailer with ample stock to a retailer that has reached stockout (Lee, 1987). PLT reduces risk by redistributing stock between retailers that anticipate stockout before the realization of customer demands (Tagaras, 1999). In short, ELT responds to stockout while PLT reduces the risk of future stockout.

Lee (1987) presented a model that allows ELT between local warehouses that are part of a group. If a local warehouse cannot satisfy customer demands with its stock on hand, ELT is used to fill the demands from a

warehouse in the same group that has enough stock on hand. If ELT is impossible due to group-wide stockout, the goods in demand will be backordered. Lee (1987) derived expressions that approximate the fractions of demands that can be satisfied by stock on hand, ELT, and backordering, and in doing so, proved that applying lateral transshipment reduces total cost. Tagaras (1999) proposed random and risk-balancing policies. He organized many scenarios and evaluated them by the total expected cost and fulfill rate. Axäster (1990) analyzed a system similar to that of Lee (1987), but with the modification of assuming that warehouses within each group are not identical. Axäster (1990) derived steady-state probability by assuming exponentially distributed replenishment time. Analytical results were compared with simulation results to show that, in the case of non-identical warehouses, the proposed model gives better results. The redistribution policy proposed by Bertrand and Bookbinder (1998) adjusted stock to achieve equal marginal cost over all retailers just before the replenishment period. This method has the disadvantage of not being able to respond to stockout before redistribution because the redistribution policy is only performed at the end of the replenishment period.

Lateral transshipment based on availability (TBA), one of two kinds of policies proposed by Banerjee et al. (2003), redistributes stock to retailers with less than desirable levels until all stock is depleted. However, this policy is problematic when desired stock levels are determined incorrectly. lateral transshipment for inventory equalization (TIE) redistributes stock to match the ratio of average demand of each retailer to that of the whole retailer whenever there are retailers with less than desirable stock levels. This policy cannot respond to stockout after redistribution, because redistribution is performed once in every replenishment cycle. Previous policies have the disadvantage of not being able to respond to stockout before or after redistribution, and they cannot appropriately determine desired stock levels.

Some researchers have attempted heuristic approximations and/or simulation approaches to overcome the complexity involved with analytical modeling and solving multi-echelon supply chain problems (Banerjee et al., 2003; Burton and Banerjee, 2005). Finding exact solutions to such problems can be very time-consuming and difficult to implement in practice (Axäster, 2003). Moreover, transshipment is associated with secondary effects including changes to the time and size of one or more future replenishments at the location releasing the transshipment (Minner et al., 2003). Therefore, we select the simulation approach to find the best lateral transshipment policy.

Entire lateral transshipment policies may be problematic if retailers' on-hand stock is used as their only criteria. Definitions of service levels from previous studies are shown in Table 1. Because it is measured considering future demands, current stock quantity, and the degree of stockout, the service level proposed in this study can be used as criteria to evaluate the performance of lateral transshipment. In this paper, we define the service level to coincide with a newly proposed lateral transshipment policy, and we apply the service level to determine the desired stock level.

In this study, we propose service level adjustment (SLA) as a new lateral transshipment policy to effectively deal with retailer demands. This policy reduces risk by forecasting stockout in advance and efficiently responding to actual stockout by combining ELT with PLT. Moreover, we define the new service level as the performance criteria of SLA.

In Section 2, we propose SLA using the service level in the remaining period SLRP as the service level. In Section 3, we compare the total cost of SLA with that of previous policies by simulating demand changes and various costs. Finally, we provide conclusions in Section 4.

Table 1  
Definitions of service level

|                                                    |                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cycle-service level (Lee and Larry, 2002)          | The desired probability of not running out of stock in any one ordering cycle                                                                                                                                                                                                                                                                                      |
| Customer service level (Yan et al., 2003)          | A percentage of the total demand met                                                                                                                                                                                                                                                                                                                               |
| Customer service level (Biswas and Narahari, 2004) | (1) Order fill rate: the fraction of customer demands that are met from stock<br>(2) Probability of on-time delivery: the fraction of customer orders that are fulfilled on-time                                                                                                                                                                                   |
| Service level (Sürle and Wagner, 2002)             | (1) $\alpha$ -service level : the probability that an incoming order can be fulfilled completely from stock<br>(2) $\beta$ -service level : the proportion on incoming order quantities that can be fulfilled from inventory on-hand<br>(3) $\gamma$ -service level: $1 - \frac{\text{mean demand not fulfilled by end of period}}{\text{mean demand per period}}$ |

## 2. Service level adjustment

This study focuses on an item that has high stockout cost in a two-echelon supply chain involving a supplier and multiple retailers (Fig. 1). Demand for the item at each retailer in each period is stochastic, stationary, normally distributed and independent of the demand of other retailers. The inventory policy is the  $(R, S)$  policy. Ordinarily, retailers are located much closer to each other than to their supplier, and customer demands are fulfilled by on-hand stock. At the end of each review period, all retailers review their level of inventory and order from one supplier. The ordered quantity is determined to bring the inventory positions up to order-up-to level. The equation to determine order-up-to level is presented in Table 2. Lead time is necessary for retailers to receive ordered products. For this study, the lead time required to replenish each retailer is assumed to be equal and deterministic. The lead time between retailers is extremely short compared to that between the supplier and each retailer because lateral transshipment is executed by fast transportation between nearby retailers. If a retailer does not have enough stock to react to future demands before receiving ordered products, they satisfy emergency demands by having products transshipped from retailers with ample stock.

In this study, we propose a new lateral transshipment policy that uses the SLRP based on the concept of safety stock. The SLRP in Fig. 2 indicates the probability that stockout will not occur for retailer  $i$  before the reorder point after period  $t$ , which is the remaining period  $RP(t)$ .  $RP(t)$  is expressed in expression (1), and the mathematical definition of SLRP is represented in expression (2).

$$RP(t) = R - t, \quad (1)$$

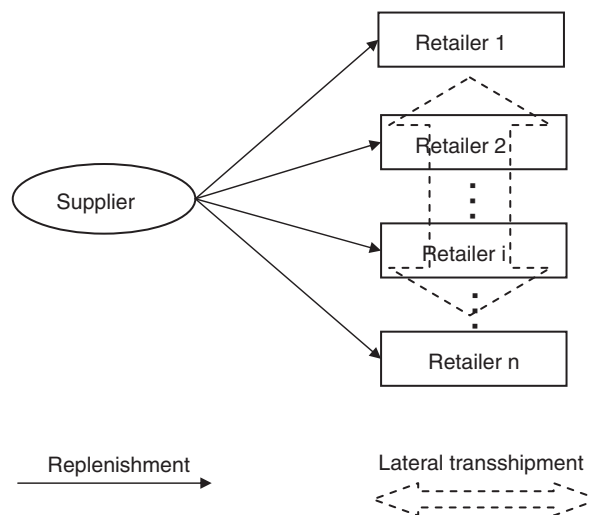


Fig. 1. Lateral transshipment in the supply chain.

Table 2  
Input parameters

|                      |                                                |
|----------------------|------------------------------------------------|
| $N$                  | 10                                             |
| $R$                  | 20 (days)                                      |
| $L$                  | 7 (days)                                       |
| $S_i$ (Ballou, 1999) | $\mu_i(R + L) + z_{0.7}(\sigma_i\sqrt{R + L})$ |
| CH                   | 1, 5, 10                                       |
| CP                   | 4, 40, 80                                      |
| CTQ                  | 1, 10, 20                                      |
| CTO                  | 1, 10, 20                                      |

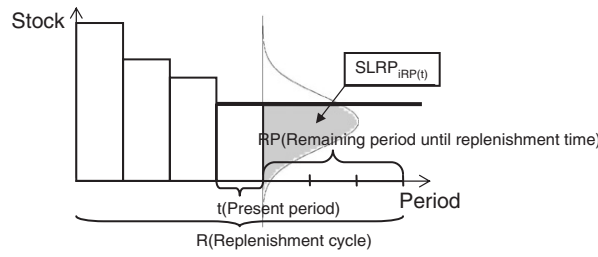


Fig. 2. SLRP.

$$\begin{aligned}
 \text{SLRP}_{iRP(t)} &= P(\text{SD}_{iRP(t)} < \text{INV}_{it}) \\
 &= P(\text{SD}_{iRP(t)} < \text{OH}_{it} - \text{SO}_{it}) \\
 &= \Phi\left(\frac{(\text{OH}_{it} - \text{SO}_{it}) - \mu(\text{SD}_{iRP(t)})}{\sigma(\text{SD}_{iRP(t)})}\right).
 \end{aligned} \tag{2}$$

In this paper, we propose SLA based on the SLRP that is defined in advance. This policy determines the quantity of the lateral transshipment at the end of the unit period. We introduce upper, lower, and target service levels to determine quantity for lateral transshipment among retailers. These levels are specific values equal to the probability that stockout does not occur in a certain period for each retailer. These values are treated as the target of operation by retailers. Therefore, they are fixed before operations begin. The upper service level indicates that a retailer has enough stock to satisfy customer demands. If any stock remains after satisfying the upper level, this stock is supplied to retailers with stockout or low stock. The target and lower service levels indicate the maximum and minimum levels that retailers with stockout or low stock can receive. When there is enough stock available among all retailers, some retailers will receive products to satisfy the target level. But some will only receive enough stock to meet the lower level, if there is not enough stock. After all, even the lower level meets customer satisfaction. Forecasted demands, lost sales, and various costs including those associated with inventory can be used to determine service levels. However, it is very important to also consider qualitative factors like strategy and business environment because customer satisfaction should be considered during strategy design. Therefore, manager judgment is often used to determine or adjust the level of customer satisfaction. Results based on judgment are often recognized as being more accurate than those obtained by certain analytic solutions. The subjective nature of factors affecting customer satisfaction makes the determination of service levels difficult. Therefore, it is assumed that each service level is strategically determined by managers and partners.

SLA operates based on several replenishment scenarios. When a retailer is not replenished by a supplier in the current period (Fig. 3), upper and lower levels of the SLRP are compared with the measured SLRP for each retailer. In this case, retailer  $j$  with high stock level redistributes available stock to the target level of the  $\text{SLRP}(\beta)$  to retailer  $k$  with a low stock level. Next, when a retailer is replenished by a supplier in the current period (Fig. 4), the stock of each retailer is measured. Then, retailers with available stock redistribute their stock as much as the quantity for stockout to retailers with stockout.

SLA for the unit period can be expressed by the following steps.

Step 0. Each retailer uses expressions (3) and (4) to satisfy demands for their stock.

$$\text{OH}_{it} = \begin{cases} \text{OH}_{it-1} - D_{it}, & \text{OH}_{it-1} \geq D_{it}, \\ 0, & \text{OH}_{it-1} < D_{it}, \end{cases} \tag{3}$$

$$\text{SO}_{it} = \begin{cases} 0, & \text{OH}_{it-1} \geq D_{it}, \\ D_{it} - \text{OH}_{it-1}, & \text{OH}_{it-1} < D_{it}. \end{cases} \tag{4}$$

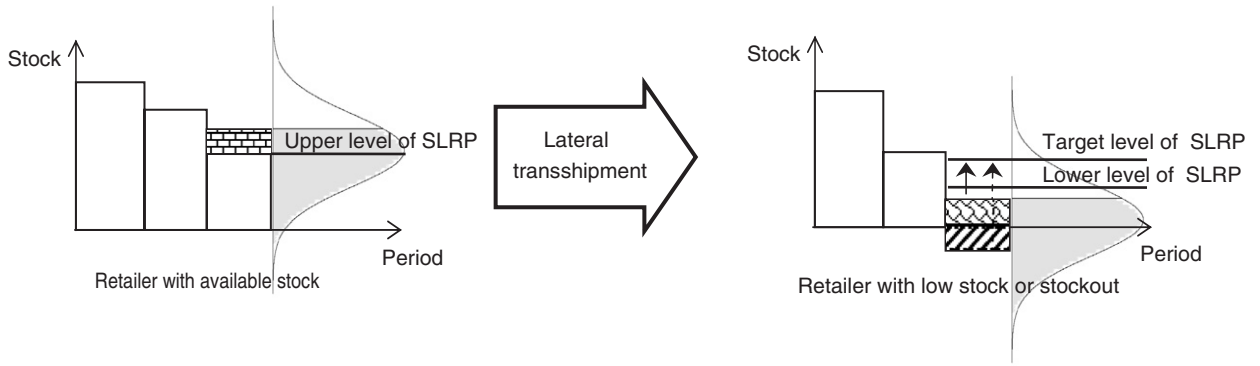


Fig. 3. SLA: lateral transshipment for a period without replenishment.

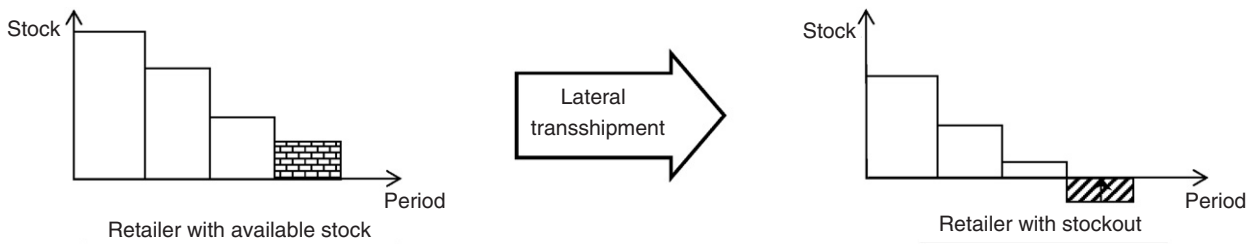


Fig. 4. SLA: lateral transshipment for a period with replenishment.

### Step 1. Searching to perform lateral transshipment

#### 1.1. In case of no replenishment in the current period

- Use the upper and lower levels of the SLRP to search for retailer  $j$  with a high stock level that satisfies expression (5) and for retailer  $k$  with a low stock level that satisfies expression (6).

$$\text{int}(\mu(\text{SD}_{i \text{ RP}(t)}) + z_\alpha \sigma(\text{SD}_{i \text{ RP}(t)}) + 0.5) < \text{OH}_{it} - \text{SO}_{it} \quad (5)$$

$$\text{int}(\mu(\text{SD}_{i \text{ RP}(t)}) + z_\gamma \sigma(\text{SD}_{i \text{ RP}(t)}) + 0.5) \geq \text{OH}_{it} - \text{SO}_{it} \quad (6)$$

#### 1.2. In case replenishment in the current period

- Search for retailer  $j$  with stock that satisfies expression (7) and for retailer  $k$  with stockout that satisfies expression (8).

$$\text{OH}_{it} > 0, \quad (7)$$

$$\text{SO}_{it} > 0. \quad (8)$$

If the set of retailers  $J$  and  $K$  exists, go to step 2. Otherwise, go to step 0.

Step 2. Use expressions (9)–(11) to calculate quantity for lateral transshipment between retailers in the unit period.

$$\text{RS}_{kt} = \begin{cases} \max(\text{int}(\mu(\text{SD}_{k \text{ RP}(t)}) + z_\beta \sigma(\text{SD}_{k \text{ RP}(t)}) + 0.5), 0) - (\text{OH}_{kt} - \text{SO}_{kt}), & \text{RP}(t) \geq 1, \\ \text{SO}_{kt} & \text{RP}(t) = 0, \end{cases} \quad (9)$$

$$ERS_{kt} = \begin{cases} \max(\text{int}(\mu(\text{SD}_k \text{RP}(t)) + z_\beta \sigma(\text{SD}_k \text{RP}(t)) + 0.5), 0) - \text{OH}_{kt}, & \text{RP}(t) \geq 1, \\ 0, & \text{RP}(t) = 0, \end{cases} \quad (10)$$

$$AS_{jt} = \begin{cases} \text{OH}_{jt} - \max(\text{int}(\mu(\text{SD}_j \text{RP}(t)) + z_\alpha \sigma(\text{SD}_j \text{RP}(t)) + 0.5), 0), & \text{RP}(t) \geq 1, \\ \text{OH}_{jt}, & \text{RP}(t) = 0. \end{cases} \quad (11)$$

Step 3. If the quantity of total available stock is smaller than the quantity of total required stock or stockout, adjust quantity for stock.

3.1. If the quantity of total available stock is between the quantity of total required stock and total stockout, adjust the required quantity of stock by following expression (12).

$$RS_{kt} = \text{int} \left( \text{SO}_{kt} + ERS_{kt} \left( \sum_j AS_{jt} - \sum_k \text{SO}_{kt} \right) / \sum_k ERS_{kt} \right). \quad (12)$$

3.2. If the quantity of total available stock is smaller than the quantity of total stockout, follow expression (13) to adjust required stock quantities.

$$RS_{kt} = \text{int} \left( \text{SO}_{kt} \left( \sum_j AS_{jt} \right) / \sum_k \text{SO}_{kt} \right). \quad (13)$$

Step 4. Retailer  $j$  with the highest available stock redistributes stock to retailer  $k$  with the highest required stock in the current period.

4.1. Follow expression (14) to determine the quantity for lateral transshipment. Update the quantity of required and available stock following expressions (15) and (16).

$$X_{jkt} = \min(\max(RS_{kt}), \max(AS_{jt})), \quad (14)$$

$$RS_{kt} \leftarrow RS_{kt} - X_{jkt}, \quad (15)$$

$$AS_{jt} \leftarrow AS_{jt} - X_{jkt}. \quad (16)$$

Retailer  $j$  redistributes stock with  $X_{jkt}$  to retailer  $k$ . Then, on-hand stock and quantity for stockout of retailer  $k$  are updated by expressions (17)–(18). Eq. (19) means on-hand stock of retailer  $j$ .

$$\text{OH}_{kt} \leftarrow \begin{cases} \text{OH}_{kt} + (X_{jkt} - \text{SO}_{kt}), & X_{jkt} \geq \text{SO}_{kt}, \\ 0, & X_{jkt} < \text{SO}_{kt}, \end{cases} \quad (17)$$

$$\text{SO}_{kt} \leftarrow \begin{cases} 0, & X_{jkt} \geq \text{SO}_{kt}, \\ \text{SO}_{kt} - X_{jkt}, & X_{jkt} < \text{SO}_{kt}, \end{cases} \quad (18)$$

$$\text{OH}_{jt} \leftarrow \text{OH}_{jt} - X_{jkt}. \quad (19)$$

Step 5. If the quantity of total required stock satisfies expression (20), go to step 6. Otherwise, go to step 4.

$$\sum_k RS_{kt} = 0, \quad (20)$$

Step 6. If there is no replenishment in the current period, each retailer gives an order of  $Q_{it}$  products to the supplier.  $Q_{it}$  is calculated by following expression (21). Then, go to step 0.

$$Q_{it} = S_i - OH_{it}. \quad (21)$$

Step 7. If there is replenishment in the current period, each retailer receives  $Q_{it-L}$  products from the supplier. Therefore, on-hand stock of retailer  $i$  is updated by following expression (22). Then, go to step 0.

$$OH_{it} \leftarrow OH_{it} + Q_{i\ t-L}. \quad (22)$$

### 3. Numerical example

In this study, we measure SLA performance as total cost of the supply chain. The total cost is made up of holding, backorder, transportation, and ordering costs. Total cost can be depicted as in expression (23).

$$TC = \sum_i \left\{ \sum_i (CH \cdot OH_{it} + CP \cdot BO_{it}) + \sum_j \sum_k (CTQ \cdot X_{jkt} + CTO \cdot e_{jkt}) \right\}. \quad (23)$$

We prove the efficiency of the proposed policy by comparing results of simulation experiments conducted with SLA and previous policies such as TBA, TIE, and NLS (no lateral shipments) by Banerjee et al. (2003). Parameters of the numerical example are expressed in Table 2. The demand for retailers in each period is depicted as in Table 3.

We used the upper ( $\alpha$ ), target ( $\beta$ ), and lower ( $\gamma$ ) levels of SLRP to express various environments. The eight scenarios that were used are depicted in Table 4.

Simulation experiments were executed for 660 days considering steady-state (500 days) and warm-up (160 days) periods, which were established by Welch's procedure (Law and Kelton, 2000). Fig. 5 presents the warm-up period. For eight scenarios, we selected the SLA (max) with the greatest average total cost and the SLA (min) with the smallest average total cost. Minimum and maximum SLA policies were compared with TEA, TIE and NLS policies.

Fig. 6 addresses the total cost of SLA that is not dependent on certain scenarios. The determination of upper, target and lower levels of the optimal SLA is shown to hardly affect the total cost of the policy. In most cases, the SLA policy has lower total cost than any of the previously mentioned policies. Fig. 7 shows cost to be lower when high demand variation exists. Fig. 8 shows cost to be lower with increasing backorder cost, as stockout rarely occurred when the SLA policy was used. Fig. 9 shows cost to be lower with increasing ordering

Table 3  
Demands for retailers

| Retailer | A set: $N(\mu_i, \sigma_i^2)$ |                    | B set: $N(\mu_i, \sigma_i^2)$ |                    |
|----------|-------------------------------|--------------------|-------------------------------|--------------------|
|          | Mean                          | Standard deviation | Mean                          | Standard deviation |
| 1        | 10.5698                       | 0.868752           | 10.5698                       | 21.3144            |
| 2        | 11.6087                       | 1.10897            | 11.6087                       | 1.10964            |
| 3        | 11.8455                       | 3.43202            | 11.8455                       | 19.7631            |
| 4        | 12.0229                       | 2.02251            | 12.0229                       | 27.7476            |
| 5        | 13.1187                       | 2.27434            | 13.1187                       | 10.2159            |
| 6        | 11.3236                       | 4.82258            | 11.3236                       | 8.07008            |
| 7        | 10.0833                       | 3.03091            | 10.0833                       | 3.76491            |
| 8        | 14.7457                       | 0.466447           | 14.7457                       | 6.766              |
| 9        | 11.1748                       | 4.57645            | 11.1748                       | 16.2423            |
| 10       | 10.3163                       | 1.4491             | 10.3163                       | 13.5748            |

Table 4  
Upper, target and lower levels of SLRP

|          |                                                      |
|----------|------------------------------------------------------|
| $\alpha$ | 0.5, 0.7                                             |
| $\beta$  | 0.3, 0.5                                             |
| $\gamma$ | 0.1, 0.2                                             |
|          | $\gamma < \beta \leq \alpha, 0 < \gamma, \alpha < 1$ |

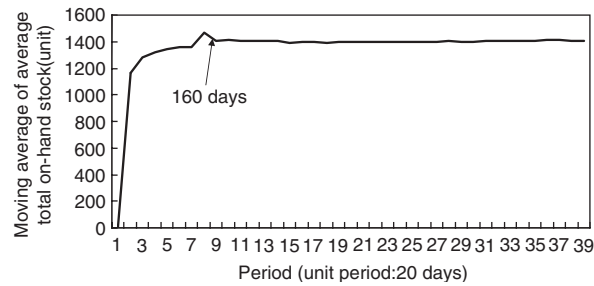


Fig. 5. The warm-up period by Welch's procedure.

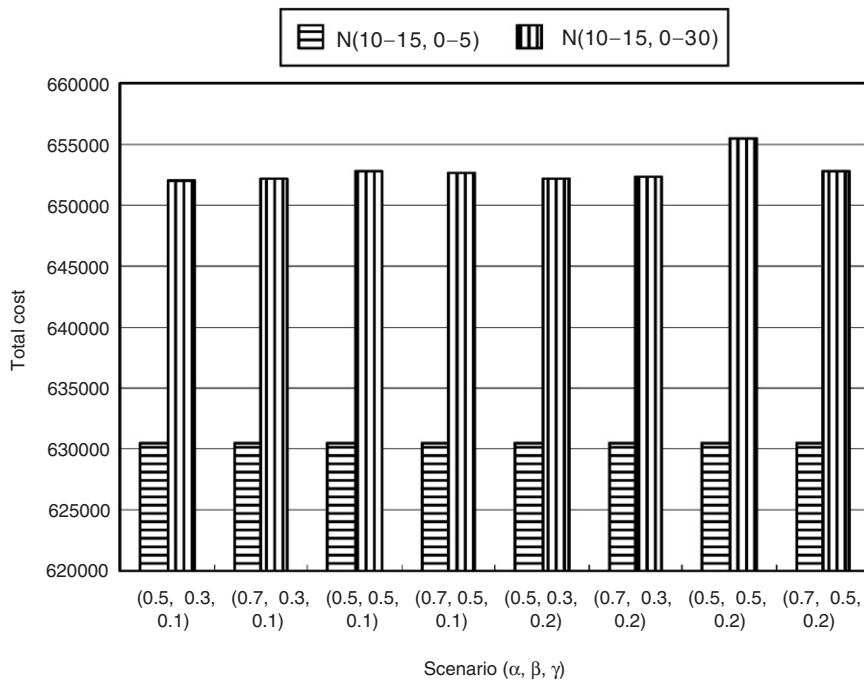


Fig. 6. Total cost by scenario.

cost. Fig. 10 shows cost to be lower with increasing holding cost, although there is little difference between policies. Fig. 11 shows that the total cost of the SLA policy is higher than that of the other policies with increasing transportation cost. This is due to the high number of transshipments needed to maximally satisfy customer demands.

Given the above, the SLA policy is more efficient than previous policies in terms of customer demand uncertainty, on-hand stock, and stockout. In cases where transportation costs between two retailers are reducible, SLA will be a useful lateral transshipment policy.

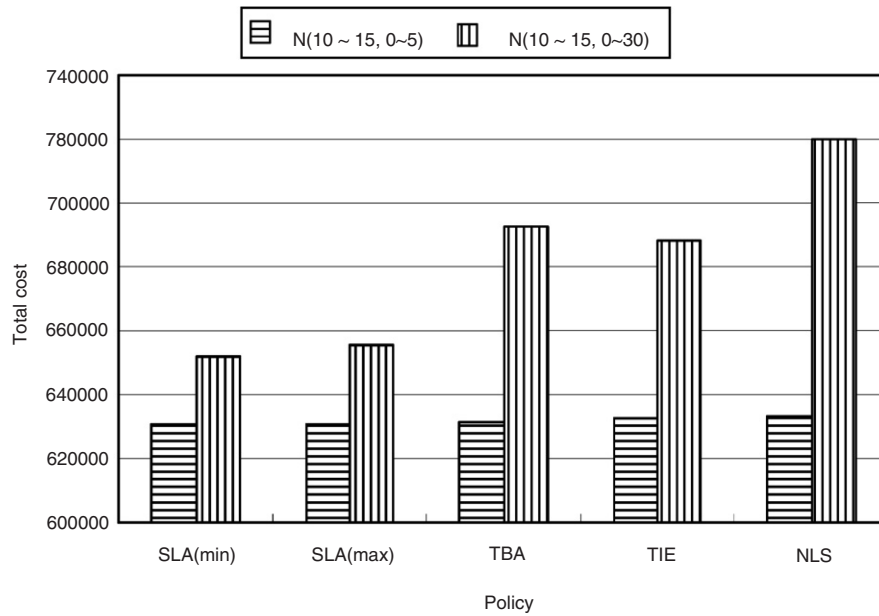


Fig. 7. Total cost by lateral transshipment policy.

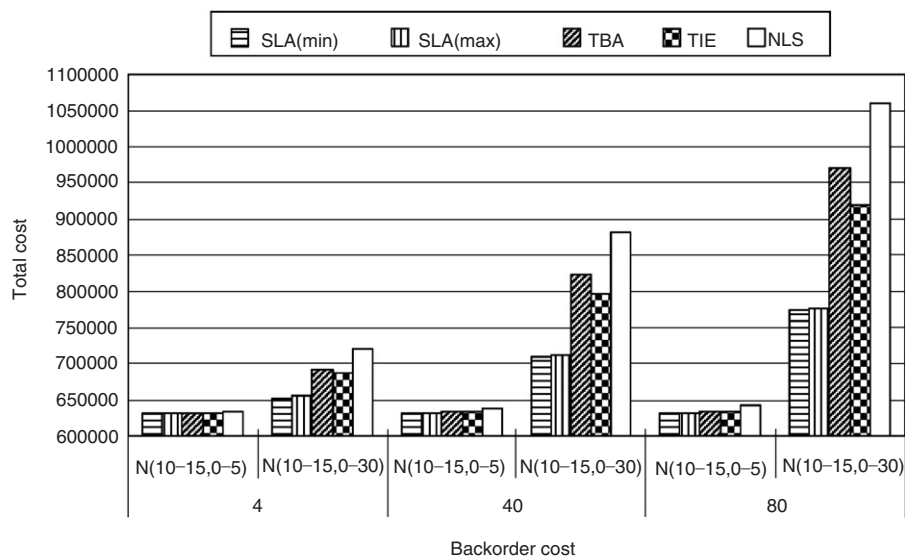


Fig. 8. Total cost by backorder cost.

#### 4. Conclusion and future work

Lateral transshipment policies can maximize customer satisfaction and reduce management costs. Various previously proposed lateral transshipment policies have the disadvantage of not suitably responding to customer demands. In this study, we defined SLRP as a service level and proposed the SLA policy to efficiently deal with demands on retailers. We showed that SLA is more effective than previous policies through simulation experiments. Because SLA concurrently considers ELT and PLT, all of the advantages of the two previous policies are realized in this paper. If transportation unit cost between two retailers is sufficiently low, SLA will be more effective than TBA, TIE, and NLS policies.

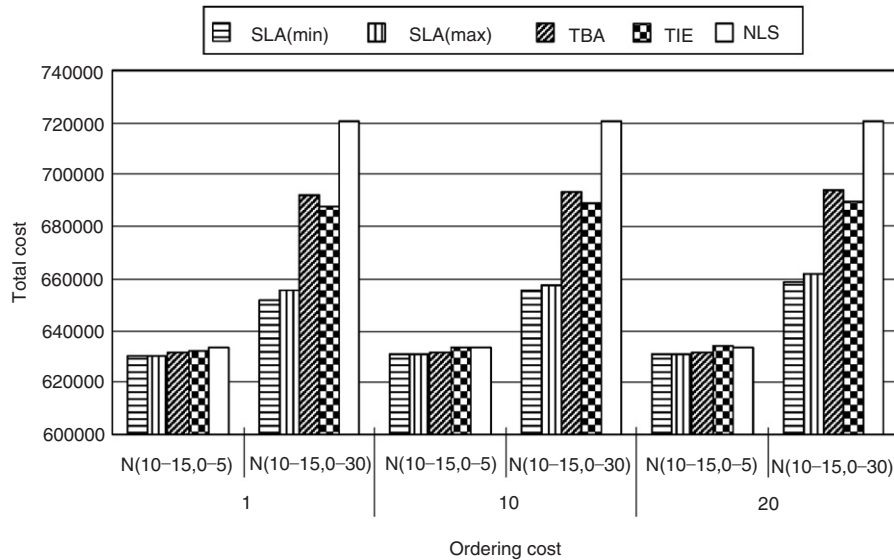


Fig. 9. Total cost by ordering cost.

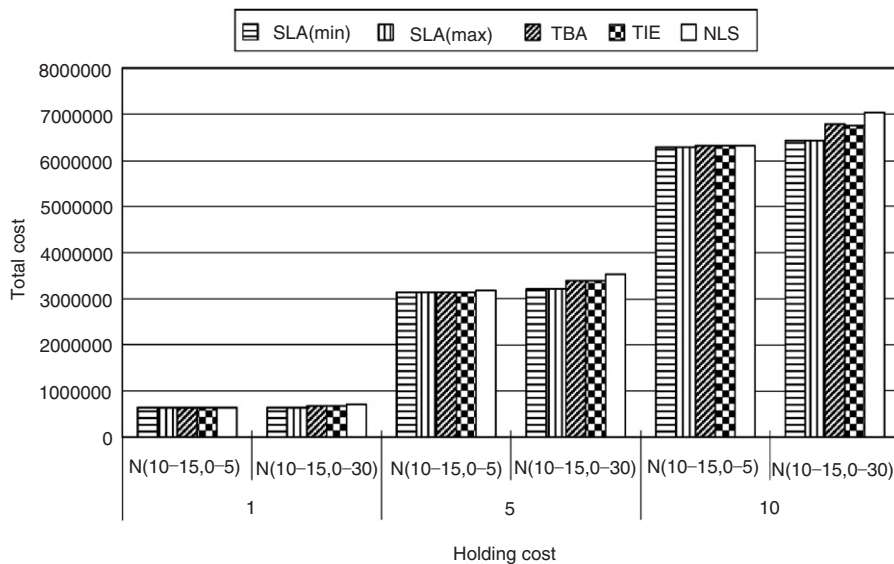


Fig. 10. Total cost by holding cost.

Under increasing transportation unit cost between retailers, SLA may not be a good policy. However, many distributors are improving the affordability of their management technology through third-party logistics (3PL), consolidated transportation, and other methods. Moreover, many logistics service providers compete for price. Therefore, transportation cost may be assuaged to overcome the drawback of SLA.

We are considering the use of a periodic review system for this study, but additional studies of lateral transshipment are needed, including those that incorporate inventory policy systems according to the characteristics of products in the supply chain.

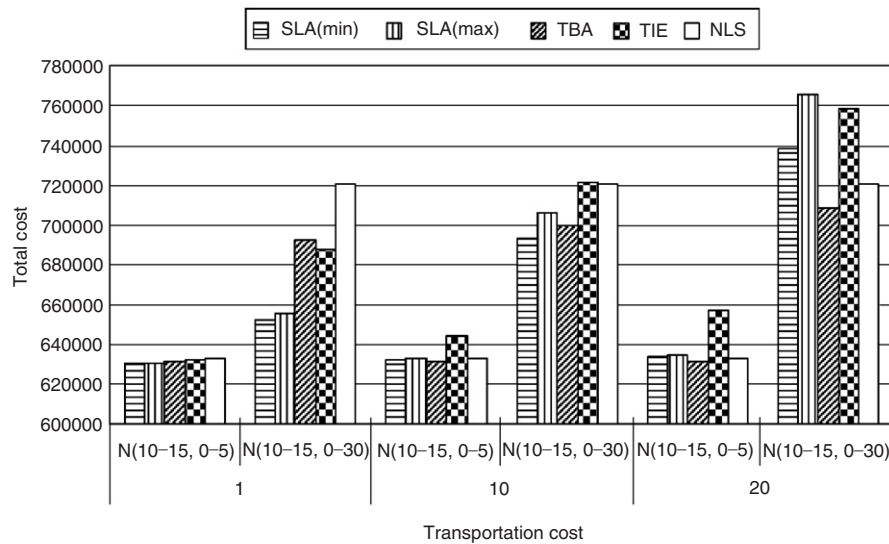


Fig. 11. Total cost by transportation cost.

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